Course: BTech

Semester: 6

Prerequisite: Fundamental of Programming, Computer Network

Rationale : Cyber security is the application of technologies, processes, and controls to protect systems, networks, programs,devices, and data from cyber-attacks. It aims to reduce the risk of cyber-attacks and protect against the unauthorized exploitation of systems, networks, and technologies.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	0	3	20	20	-	60	-	100

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content **W** - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Information Security** Introduction to information system, Types of information Systems, Development of Information Systems, Introduction to Information Security, Need for Information Security, Threats to Information Systems, Information Assurance, Cyber Security and Security Risk Analysis.	**15**	**8**
2	**Systems Vulnerability Scanning** Overview of vulnerability scanning, Open Port/Service Identification, Banner/ Version Check, Traffic Probe, Vulnerability Probe, Vulnerability Examples. Networks Vulnerability Scanning - Netcat, Understanding Port and Services tools, Network Reconnaissance–Nmap. Network Sniffers and Injection tools–Wireshark.	**25**	**13**
3	**Network Defense tools** Firewalls and Packet Filters: Firewall Basics, Packet Filter Vs Firewall, Firewall Protects a Network, Packet Characteristic to Filter, Stateless Vs Stateful Firewalls, Network Address Translation(NAT) and Port Forwarding, the basic of Virtual Private Networks, Linux Firewall, Windows Firewall, Snort: Introduction Detection System	**20**	**11**
4	**Introduction to Cyber Crime and law** Cyber Crimes, Types of Cybercrime, Hacking, Attack vectors, Cyberspace and Criminal Behavior, Clarification of Terms, Traditional Problems Associated with Computer Crime, Introduction to Incident Response, Digital Forensics ,Computer Language, Network Language, Realms of the Cyber world, A Brief History of the Internet, Recognizing and Defining Computer Crime, Contemporary Crimes, Computers as Targets, Contaminants and Destruction of Data, Indian ITACT 2000.	**20**	**7**
5	**Introduction to Cyber Crime Investigation** Firewalls and Packet Filters, password Cracking, Key loggers and Spyware, Virus And Warms, Trojan and backdoors, Steganography, DOS and DDOS attack, SQL injection, Buffer Overflow, Attack on wireless Networks.	**20**	**6**
Total		**100**	**45**

Reference Books	
1.	**Cryptography and Network Security (TextBook)** By William Stallings | Pearson Education
2.	**Cryptography and Network Security (TextBook)** By V.K. Jain | Khanna Publishing House
3.	**Information and Cyber Security (TextBook)** By Gupta Sarika | Khanna Publishing House
4.	**Cryptography and Network Security (TextBook)** By Atul Kahate, TMH
5.	**Cryptography and Information Security (TextBook)** By V.K. Pachghare | PHI Learning
6.	**Anti-Hacker Tool Kit** By Mike Shema | McGrawHill
7.	**Cyber Security understanding Cyber Crimes, Computer forensics and Legal Perspectives** By Nina Godbole and Sunit Belapure | WILEY

Course Outcome

After Learning the Course the students shall be able to:

1. Explain the features and characteristics of the Linux Operating System and Windows Operating System.
2. Apply network monitoring tools to identify attacks against network protocols and services.
3. Apply various methods to prevent malicious access to computer networks, hosts, and data.
4. Explain how to investigate endpoint vulnerabilities and attacks.
5. Analyze network intrusion data to verify potential exploits.
6. Apply incident response models to manage network security incidents